

A Joint Multilingual Framework for Hope Speech Detection in Bengali and English

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Presentation Outline

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Introduction

- Hope Speech Detection (HSD) is a technique that analyzes textual content to identify supportive and encouraging communication online.
- Focus on analyzing **Bengali & English** social media text. A specialized Hope Speech Detection system that can understand Bengali tweets and classify them as Hope Speech (1) or Non-Hope Speech (0).
- **Goal:** To build an AI-based system that can automatically detect hope speech in multilingual social media, promoting positivity, diversity, and peace online.

❏ Motivation

AI Knows Hates, We're Teaching It Hope

- Popular NLP systems detect hate speech in **English, Spanish & Russian** but not in **Bengali**.
- Existing research highlights negativity online but fails to capture the **supportive and cultural** context of multilingual **social media**.
- **We're building an AI that doesn't just filter—it amplifies positivity and hope in multilingual voices.**

❑ Objective

- ❑ Develop a **novel HSD system** capable of classifying Bengali & English social media text into Hope Speech (1) and Non-Hope Speech (0).
- ❑ Construct the **existing first BongHope & HopeEDI**, consisting of **2,626 & 28,450 manually annotated tweets & youtube** collected between 2021– till now with clear Hope Speech (1) vs Non-Hope Speech (0) labels.
- ❑ Ensure annotation reliability using **Fleiss' Kappa & Krippendorff's Alpha** and validate corpus balance across categories.
- ❑ Design and **benchmark computational models**, including traditional classifiers and transformer-based architectures, for accurate and robust Hope Speech detection in multilingual text.

❏ Literature Review

Existing Works

- **NLP on social media** → hate speech, emotion, humour detection.
- **Some datasets:** Mostly focused on → negativity, not hope.

Hope Speech Datasets

- **Palakodety (2019)** → English YouTube comments.
- **Chakravarthi (2020)** → HopeEDI (English, Tamil, Malayalam).
- **Tanusree (2023)** → BongHope Social Media tweets (Bangla).

Literature Review

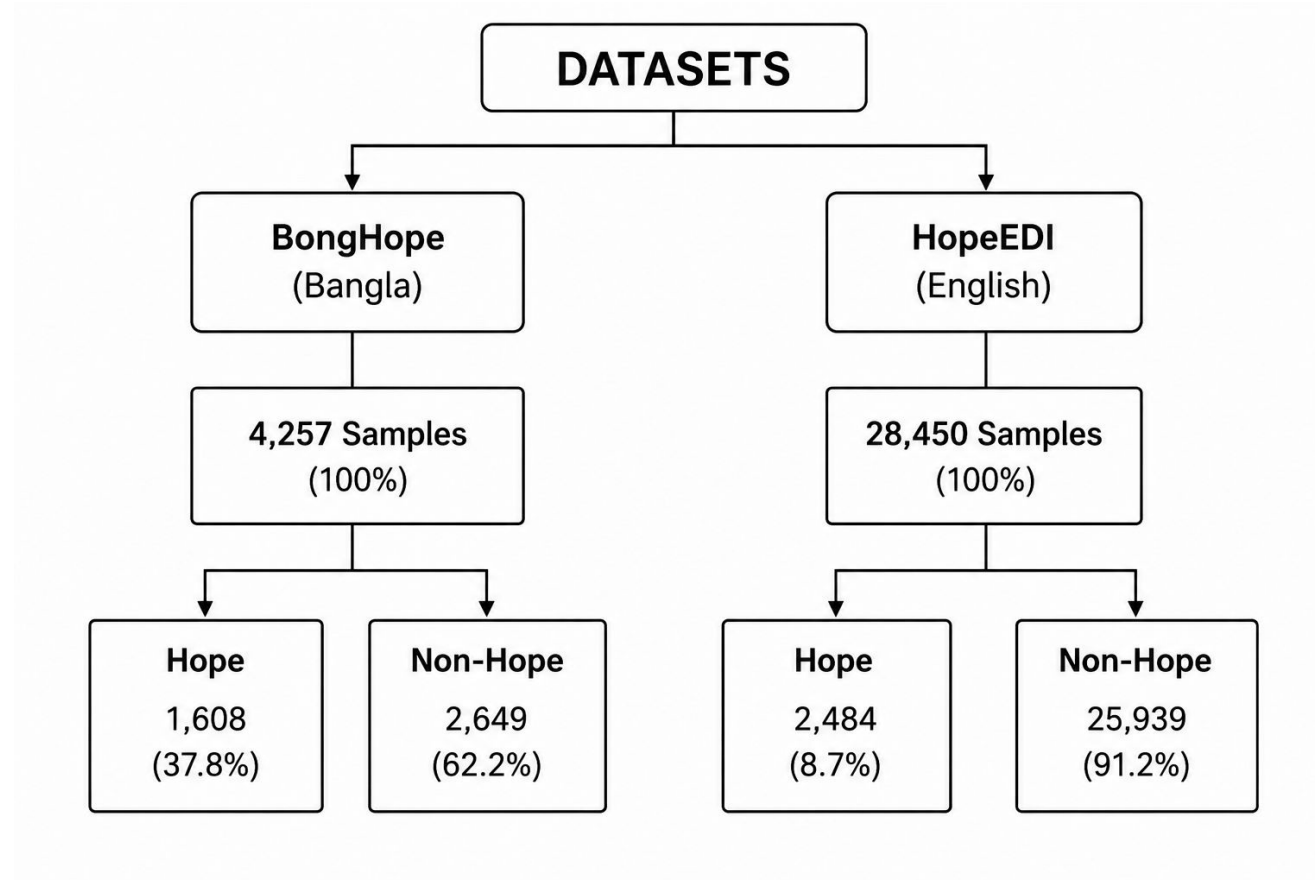
Limitations in Current Works

- Imbalance dataset for **Bengali & English** Hope Speech.
- Existing models focus on **hate/abuse** detection, not hope.
- Most research is limited to **Tamil, Malayalam, Kannada**.
- Most works is in **English, Spanish, Russian**.

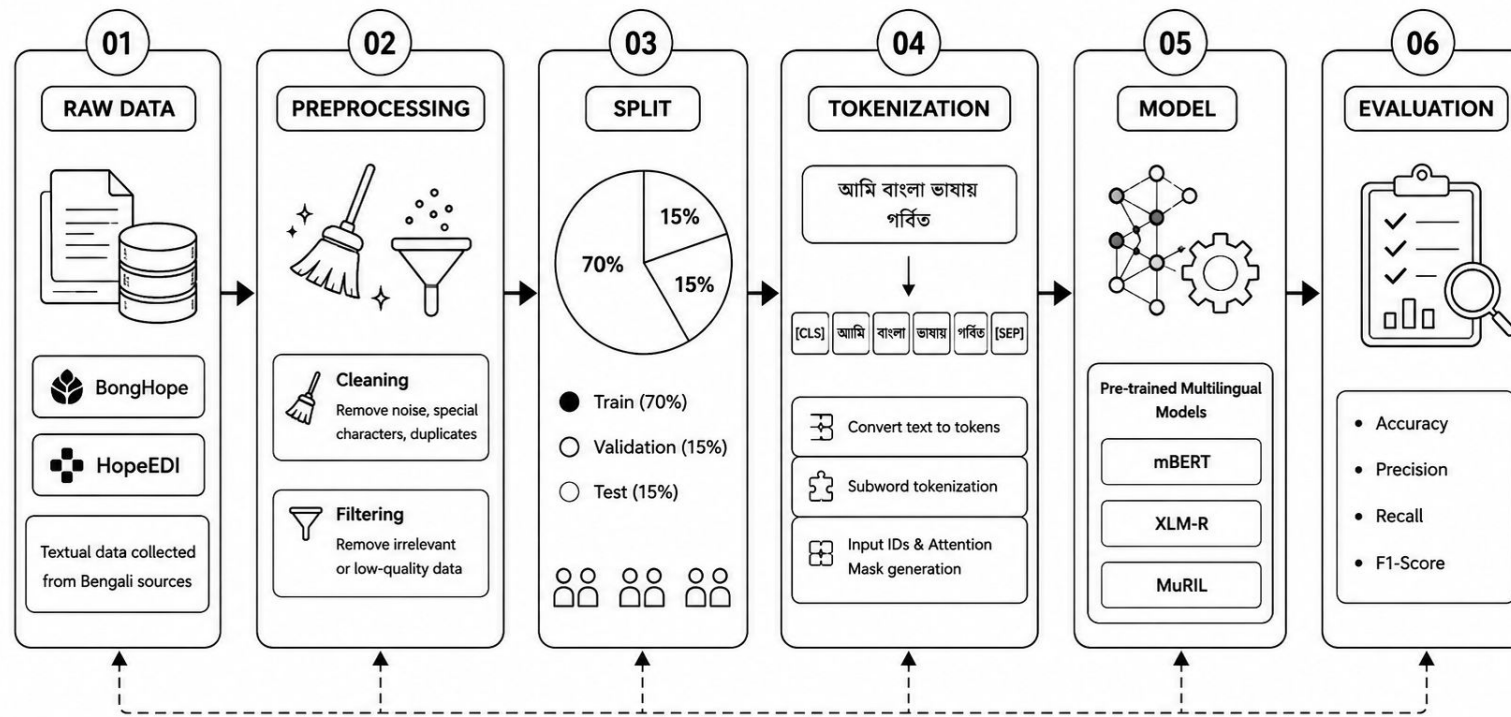
Our Focus

- Build the **existing** first **Bengali & English** Hope Speech dataset.
- Support **positivity and inclusivity** in social media discourse.
- Work with multilingual text to promote culturally aware **NLP research**.

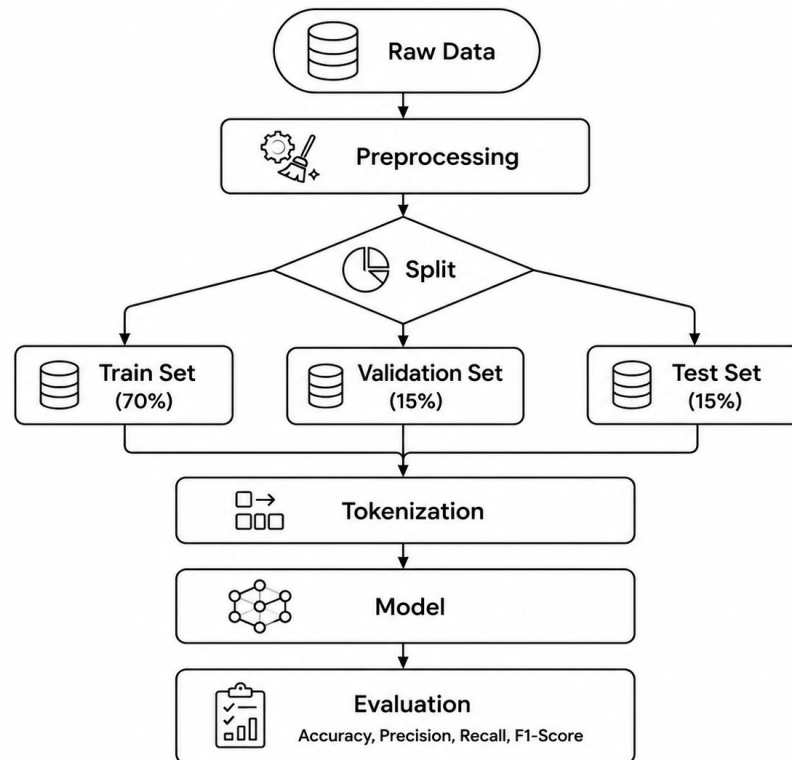
Literature Review



Methodology



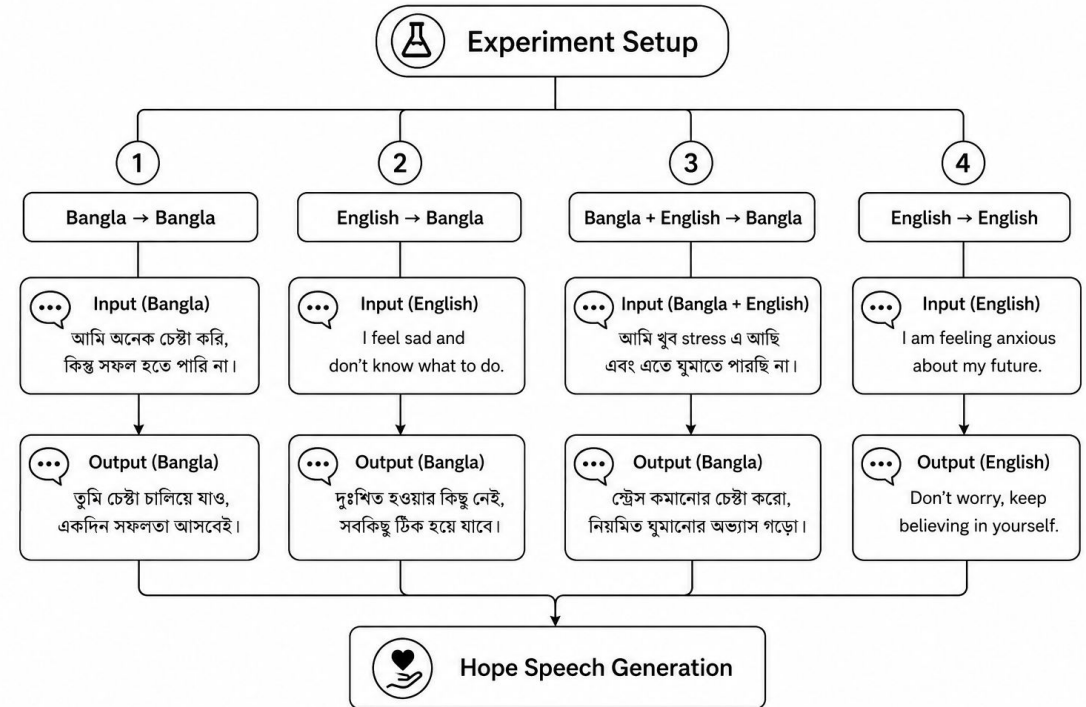
❏ Methodology



- **Raw Data:** Existing dataset BongHope & HopeEDI.
- **Preprocessing:** Dataset cleaning & filtering.
- **Split:** Train 70%, Val 15% & Test 15%
- **Tokenization:** Convert text to machine codes.
- **Model:** mBERT, XLM-R & MuRIL Transformer models.
- **Evaluation:** Accuracy, F1 (Macro), Precision, Recall

Progress Demonstration

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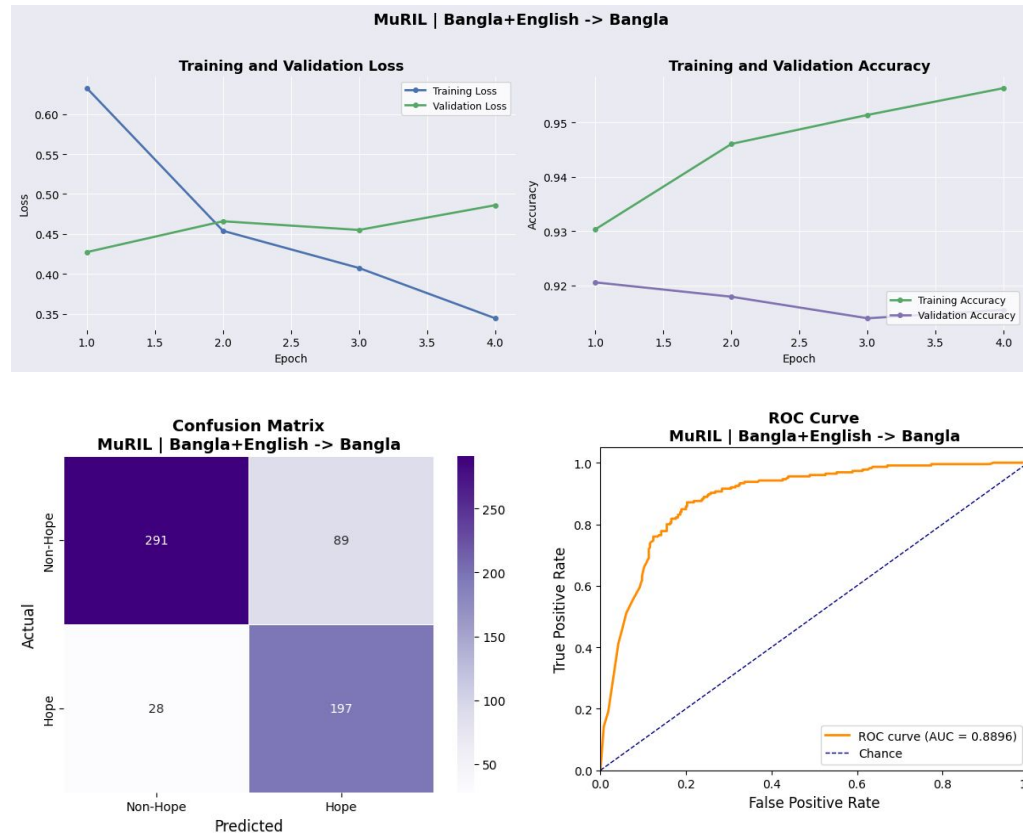


Example: আমরা একসাথে থাকলে সবকিছু সম্ভব। (Hope) And কোনো পরিবর্তন হবে না, সবই বৃথা। (Non-Hope)

Progress Demonstration

Exprements Performance Comparison						
Models	Exprements	Accuracy	F1 (Macro)	F1 (Weighted)	Precision (Macro)	Recall (Macro)
1. mBERT	English → English	0.9292	0.7799	0.9304	0.7705	0.7902
	Bangla → Bangla	0.7769	0.7541	0.7733	0.7646	0.748
	English → Bangla	0.6305	0.3948	0.4915	0.7211	0.5037
	Bangla+English → Bangla	0.7901	0.7812	0.7925	0.7772	0.7903
2. XLM-R	English → English	0.9259	0.7817	0.929	0.7603	0.8087
	Bangla → Bangla	0.8248	0.8114	0.8243	0.8133	0.8098
	English → Bangla	0.6464	0.442	0.5285	0.7713	0.5259
	Bangla+English → Bangla	0.8149	0.8091	0.8176	0.8058	0.8245
3. MuRIL	English → English	0.9251	0.7781	0.9281	0.7582	0.8029
	Bangla → Bangla	0.838	0.8257	0.8376	0.8276	0.8239
	English → Bangla	0.6836	0.5484	0.6117	0.7566	0.5817
	Bangla+English → Bangla	0.8066	0.8098	0.8097	0.8005	0.8207

Progress Demonstration



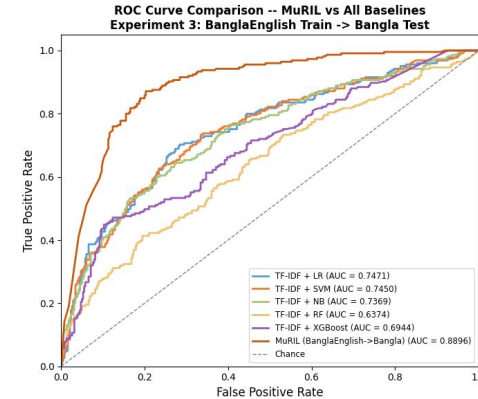
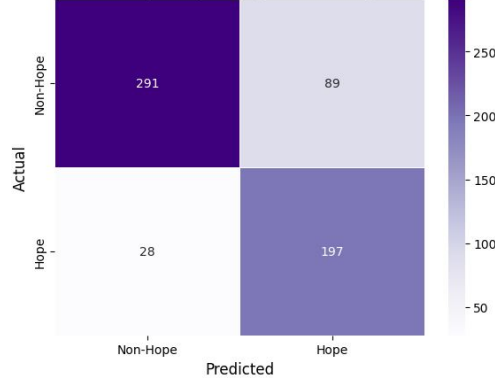
Training evaluation results of the proposed multilingual framework.

- Unstable training with smooth loss convergence
- low accuracy with strong train–validation consistency
- Top exp. result MuRIL F1 (Macro) 0.80
- Is not enough calibrated confidence and balanced class accuracy

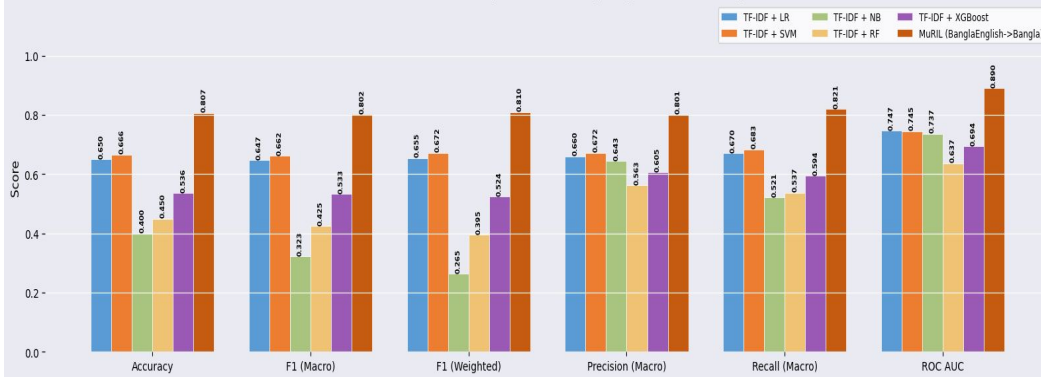


Progress Demonstration

Confusion Matrix
MuRIL | Bangla+English -> Bangla



Model Comparison -- Bangla Test Set
MuRIL vs TF-IDF Baselines (Experiment 3: BanglaEnglish Train -> Bangla Test)



- Answer options are not well balanced,
But avoiding model bias
- Total test samples : 605
- Correctly classified : 488 (80.7 %)
- False Positives (FP) : 89 (14.7 %)
- False Negatives (FN) : 28 (4.6 %)

Novelty

- **First Bengali & English** multilingual hope speech framework
- **Culturally contextual** annotations reflecting local positivity
- **multi-supported** NLP with mBERT, XLM-RoBERTa & MuRIL
- **Binary classification** (Hope / Non-Hope) with reliable annotation
- **Benchmarks comparing** traditional classifiers vs transformer models

Future Work

- Expand **dataset size** with more diverse sources (**Facebook, YouTube, LinkedIn**).
- Explore fine-grained hope **categories** (**Generalized, Realistic, Unrealistic**).
- Apply advanced **transformer models** (**GPT, mT5, Llama 3, Etc.**).
- **Multilingual** hope speech detection across **South Asian** languages.
- Integrate **multimodal** hope speech detection (**text, image, audio & video**).

Conclusion

Introduces the **first multilingual Bengali & English** hope speech framework curated from social media, combining culturally contextual annotations with transformer-based NLP models.

The system effectively classifies **Hope and Non-Hope** expressions in multilingual text with strong reliability and benchmarked performance. The results confirm the feasibility of culturally aware NLP for promoting positivity and inclusivity in low-resource languages.

Thank You